

Al-Amin Rashid Tarek

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Mirpur, Dhaka - 1216, Bangladesh

RESEARCH INTEREST

Automation and Robotics, Control Systems, Composite Materials, Additive Manufacturing

EDUCATION

- **Military Institute of Science and Technology** April 2021 - May 2025
B.SC. IN MECHANICAL ENGINEERING Dhaka, Bangladesh
 - CGPA: 3.29/4.00
 - Thesis Title: Design and Development of a Universal Robot-Assisted Rehabilitation System for Musculoskeletal Physiotherapy
- **Hazi Asmat College** 2020
HIGHER SECONDARY CERTIFICATE Dhaka, Bangladesh
 - GPA: 5.00
- **Bhairab K.B. Pilot Model High School** 2018
SECONDARY SCHOOL CERTIFICATE Dhaka, Bangladesh
 - GPA: 4.94

PROFESSIONAL EXPERIENCE

- 2025 - Present **Research Assistant**, Mechanical Engineering
Military Institute of Science and Technology, Dhaka, Bangladesh
- 2024 - 2025 **Team Co-Lead**, MIST Mongol Barota
Military Institute of Science and Technology, Dhaka, Bangladesh
- 2023 - 2024 **Mechanical Team Lead**, MIST Mongol Barota
Military Institute of Science and Technology, Dhaka, Bangladesh
- 2024 **Industrial Trainee**, Runners Limited, Dhaka, Bangladesh [Certificate]

NOTABLE PROJECTS

- **KamiKaze FPV Drone** 2025
Technology: Payload optimization, CAD modeling, Carbon fiber composite, 3D printing, Radio Communication
 - A tactical quadcopter drone system capable of carrying multiple types of munition to target and neutralize with precision and accuracy with visual control system. [\[Link\]](#)
- **AURORA X** 2025
Technology: CAD modeling, Swaybar suspension, Teensy 4.1, IoT, Wi-Fi Communication, Visual Odometry, Autonomous Traversing
 - A precision rover system was developed to perform remote manipulation, autonomous navigation, and scientific soil analysis using visual odometry and onboard instrumentation. [\[Link\]](#)
- **Multi-Site Soil Sampling and Containment Payload for Extraterrestrial Rover Applications** 2023
Technology: CAD Modeling, 3D Printing, CNC Machining, Differential Motion, Belt-Pulley Power Transmission
 - A meticulous and sophisticated rover-compatible payload system was engineered to collect soil from three subsurface sites and securely store samples, employing CAD-based design, 3D printing, CNC machining.
- **Robotic Arm** 2023
Technology: Bluetooth control, Arduino, Servo, IoT
 - A 2-DoF robotic arm built with CO₂ laser-cut plywood and servo motors, capable of lifting 250g while maintaining a low-cost design.
- **Induction Heat for Metal Heat Treatment** 2022
Technology: Electromagnetic Induction
 - A cost-effective induction heater built using SMPS parts, high-value capacitors, toroidal core, MOSFETs, and a copper coil for efficient non-contact metal heating.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

[J.1] Saqafi, A. A., Sadia, A., Tarek, A.-A. R., Alim, A. A., Tuly, F. Y., and Rahman, M. A. 2025. "Investigation of the Mechanical Properties of Repurposed-Net-Bag Reinforced Plastic." *MIST International Journal of Science and Technology* 13 (1): 59–64. doi: org/10.47981/j.mijst.13(01)2025.488(59–64).

HONORS AND AWARDS

- **Top 3 at Bangladesh BEAR Summit 2025** 2025
Bangladesh BEAR Summit [Certificate]
 - Achieved top 3 finalist in biotech category for the thesis project: *Universal Robot-Assisted Rehabilitation System for Musculoskeletal Physiotherapy*, B.Sc. final year thesis, MIST.
- **14th Place at University Rover Challenge 2025** 2025
The Mars Society, USA [Certificate]
 - Team Co Lead of MIST MONGOL BAROTA
- **2nd Runner-up in Project Showcasing in MIST Project Competition 2025** 2025
EEE Department, Military Institute of Science and Technology
 - A group project showcase
- **1st Runner-up in Anatolian Rover Challenge 2024** 2024
Space Exploration Society (UKET), Turkiye
 - Team Co Lead of MIST MONGOL BAROTA
- **5th Place in Anatolian Rover Challenge 2023** 2023
Space Exploration Society (UKET), Turkiye [Certificate]
 - Team Member of MIST MONGOL BAROTA
- **2nd Runner-up in Anatolian Rover Challenge 2022** 2022
Space Exploration Society (UKET), Turkiye [Certificate]
 - Team Member of MIST MONGOL BAROTA

CERTIFICATIONS

- **Certified SOLIDWORKS Associate, CSWA** September 28, 2024
Dassault Systèmes [Certificate]

EXTRACURRICULAR AND LEADERSHIP EXPERIENCE

Jul 2024 - May 2025	Team Co Lead , MIST Mongol Barota Military Institute of Science and Technology
Jun 2024 - May 2025	Vice Chair , ASME MIST Student Section Military Institute of Science and Technology
Jul 2023 - Jul 2024	Mechanical Team Lead , MIST Mongol Barota Military Institute of Science and Technology
Jan 2022 - Jul 2023	Mechanical Team Member , MIST Mongol Barota Military Institute of Science and Technology

REFERENCES

1. **Dr. Md Sayem Hossain Bhuiyan**
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Military Institute of Science and Technology
Email: sayem@me.mist.ac.bd
2. **Md. Anisur Rahman**
Instructor Class B, Mechanical Engineering
Military Institute of Science and Technology
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3. **Shah Md. Ahasan Siddique**
Lecturer, Mechanical Engineering
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